How Does Facial Recognition Work?

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Presentation Plan

1. What is Facial Recognition? — Definition & Real-world examples
2. The 5 Steps — From Capture to Match
3. Neural Networks & Key Vocabulary
4. Limits, Ethics & Privacy

What is Facial Recognition?

Facial recognition is a technology that identifies a person from a photo or video using artificial intelligence.

Your phone

Face ID to unlock

Airports

Passport control

Social media

Auto photo tagging

The 5 Steps of Facial Recognition

1

Capture

Camera takes a photo

2

Detect

Find the face

3

Extract

Get key features

4

Analyze

Neural network processes data

5

Match

Compare to database

Steps 1 & 2 — Capture & Detect

Step 1

Image Capture

- A camera takes a photo of a face.
- The image is made of pixels small squares of color and light.
- More pixels = better quality.
- Example: your phone camera has 12 million pixels!

Say it: "First, a camera takes a photo."

Step 2

Face Detection

- The algorithm finds the face in the image.
- It draws a bounding box around the face.
- Uses contrast, edges, and shapes to locate eyes, nose, mouth.
- Works even with multiple faces in one photo.

Say it: "Then, the algorithm finds the face."

Step 3 — Feature Extraction

The system analyzes the face
and extracts key measurements.

👁	**Eyes**	Distance between eyes
👃	**Nose**	Width and length of nose
👄	**Mouth**	Width of the mouth
📐	**Jaw**	Shape and angle of jaw
📏	**Proportions**	Distances between all points

Key vocabulary

Feature
a measurable characteristic

Extraction
finding and measuring features

Landmark
a key point on the face

Geometry
shape and spatial relationships

"Next, the system extracts key features."

Step 4 — Neural Network

What is a neural network?

- It is inspired by the human brain.
- It learns by looking at millions of face photos.
- After training, it can recognize any new face.
- The network used for faces is called a CNN — Convolutional Neural Network.

"A neural network analyzes the features."

Training data

100M+

face photos used to train the model

💡 Analogy

A neural network is like a student who has studied millions of photos. The more it studies, the better it recognizes faces.

Step 5 — Embedding & Matching

What is a face embedding?

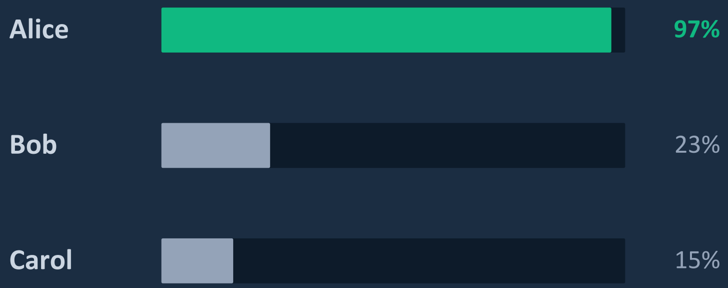
The neural network converts a face into a list of 128 numbers. This list is the face embedding ,aunique digital fingerprint.

```
[ 0.23 · 0.87 · 0.04 · 0.61 · 0.39 · 0.78 · 0.12
· 0.55 · ... ]
```

128 numbers representing one face

Matching

The system compares the new embedding to all stored embeddings in the database.



Match found! Identity confirmed.

Key Vocabulary

Pixel
FR pixel

Smallest unit of a digital image

Algorithm
FR algorithme

A set of rules for solving a problem

Bounding box
FR boîte englobante

A rectangle drawn around the detected face

Feature
FR caractéristique

A measurable aspect of the face (eye, nose…)

Neural network
FR réseau de neurones

An AI system inspired by the human brain

Embedding
FR embedding

A list of numbers representing the face

Database
FR base de données

A collection of stored face embeddings

Similarity
FR similarité

How close two embeddings are to each other

Limits & Ethics of Facial Recognition

⚠️ Algorithmic Bias

Error rates up to 34% higher for darker skin tones (MIT Media Lab, 2018). AI trained on non-diverse data performs unequally.

👀 Mass Surveillance Risk

Governments can track citizens at scae. Raises serious civil liberties concerns globally.

EU GDPR & Legal Framework

In the EU, biometric data is “special category” under GDPR. Requires explicit consent to collect or process facial data.

Conclusion & Future Perspectives

Capture → Detect →Extract →Analyze →Match

Modern systems (FaceNet, DeepFace) achieve 99%+ accuracy on benchmark datasets.

Future Directions:

- 3D facial recognition for better accuracy under different angles
- Deepfake detection — identifying AI-generated faces
- Ethical AI regulation (EU AI Act, 2024) to limit misuse

References & Sources

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